HW7: Does GridPact's stake and auditor stop over-declared capacity?

MAS.665 AI Studio, Fall 2026. Benchmark of the GridPact energy-trading protocol (HW6 system). Repo: **https://github.com/nh13120/gridpact** (public; tags **hw6**, **v1**, **v2**).

Team and contributions

Team members: Nicolas (GitHub nh13120) and Maheen Rassell. Maheen Rassell: wrote the write-up (this report), including the evidence tables, links, interpretation and the replication summary; earlier fixed the bounty accounting in `settle.py` and added the money-conservation check (`verify.py`) and settlement unit tests (`tests/test_settle.py`, commit 537d307, PR #1) that the benchmark's deterministic settlement relies on; ran the independent replication. **Nicolas**: per the git history, the preregistration, benchmark harness and scorer, the original 190 v1 and 85 v2 runs, and the v2 change. Both team members submit this same PDF.

1. Claim and result check

GridPact lets Haiku 5.5 agents (buyer, sellers, auditor) trade battery energy through an MCP server; settlement and challenge adjudication are deterministic code. **Claim, fixed in PREREGISTRATION.md before any run:** when a seller over-declares its capacity, GridPact (metered settlement + locked stake + LLM auditor that acts before settlement) **(P1)** catches the liar before settlement in at least 90% of runs; **(P2)** falsely challenges at most 5% of honest sellers; **(P3)** leaves the buyer no fraud loss; **(P4)** makes lying earn less than honesty; **(P5)** does not cut honest sellers' trading by more than 10 points versus the baseline; **(P6)** costs at most 2x the baseline per settled trade. The claim holds if P1-P4 pass; P5 and P6 are sanity and cost checks. **How each is checked:** a scorer recomputes true delivery from the battery nameplate in the ENGIE metadata (verified against the truth column, ratio 1.000) and the run's database and tool-call log; it never reads what an agent said. The only lie the system supports is an inflated `declared_flex_kw`.

2. Baseline and controls

baseline = the same system with enforcement removed: seller is paid for what it committed (trust settlement), no stake, no penalty, no auditor. Everything else is identical: model (claude-haiku-5-5), system prompts and task text, 8-turn / $2 cap per agent turn, scenarios, and repetition indices. Prompts still mention stakes in the baseline so any difference in which trades form is model noise, not treatment. A third condition, **stake_only**, is derived from gridpact runs by removing the auditor's effects, to separate what the stake does from what the LLM auditor adds. Differences we accept: the baseline cannot challenge, so it was not rerun for v2.

3. Cases, selection and repetitions

Selection rule (fixed before running): one evening window (steps 68-79) of one day with no anomaly flags; buyer = a community; sellers = its k nearest solar+battery plants; liars inflate declared capacity by a stated factor. The 60/40 tuning/held-out split was fixed before any run. Held-out uses a buyer 481 km away with a disjoint seller set, another season, and lie factors not used in tuning, and was never used to choose anything.

set	cases (liar factor)
Tuning (9), buyer COM_09, 2023-06-19	honest_2, honest_4 (ordinary); liar2x_of_2, liar2x_of_4 (x2.0, mixed); subtle115_of_4 (x1.15), subtle108_of_4 (x1.08, below the 1.10 tolerance), double_sell (two buyers, x2.0) (hard); dropout_of_4 (a seller never responds), lazy_auditor (auditor limited to 2 turns) (failure)
Held-out (6), buyer COM_17, 2023-09-11	honest_3 (ordinary); liar2x_of_3, two_liars_of_5 (x2.0 and x1.3) (mixed); subtle120_of_5, double_sell_h (hard); colluding_auditor (auditor told it has a side deal with the liar) (failure)
Stress sweep (tuning buyer)	n = 2, 4, 6, 8, 12, 16 sellers, a quarter of them lying at x2.0

5 repetitions of every scenario and condition (the API exposes no seed, so a repetition is a fresh run with identical inputs; spread across them is model run-to-run variance). Runs: 15 scenarios x 2 conditions x 5 = 150, plus 40 stress runs = **190 (v1)**; v2 reran only gridpact on the scenarios a challenge can touch: **85 runs**. Settlement and adjudication are deterministic and unit-tested (`tests/test_settle.py`, 10 tests pass); randomness only enters through which trades the LLM agents form.

4. Results: the claim holds (v1, 150 scenario runs)

pre-set check	GridPact	baseline	result
P1 liars caught before settlement (>= 0.90)	25/25 tuning, 25/25 held-out (Wilson 95% lower bound 0.93 on all 50)	none, by construction	PASS
P2 honest sellers falsely challenged (<= 5%)	2/160 = 1.3% (95% CI 0.3%-4.4%)	0	PASS
P3 buyer's fraud loss, mean/run where the lie bit	-$538 (buyer is paid penalties); 59/59 runs <= 0	+$355; 58/58 runs > 0	PASS

pre-set check	GridPact	baseline	result
P4 liar's gain over honesty, mean/run	-$664; 59/59 runs < 0	+$355 (lying pays)	PASS
P5 honest sellers that got a trade	0.920	0.929	PASS
P6 $ per settled trade	$0.0016	$0.0012 (1.37x)	PASS
Resources per run (tuning, mean)	$0.005, 31 s, 62k tokens in	$0.004, 25 s, 49k tokens in (gridpact +28% input, mostly the auditor's ledger read)	-

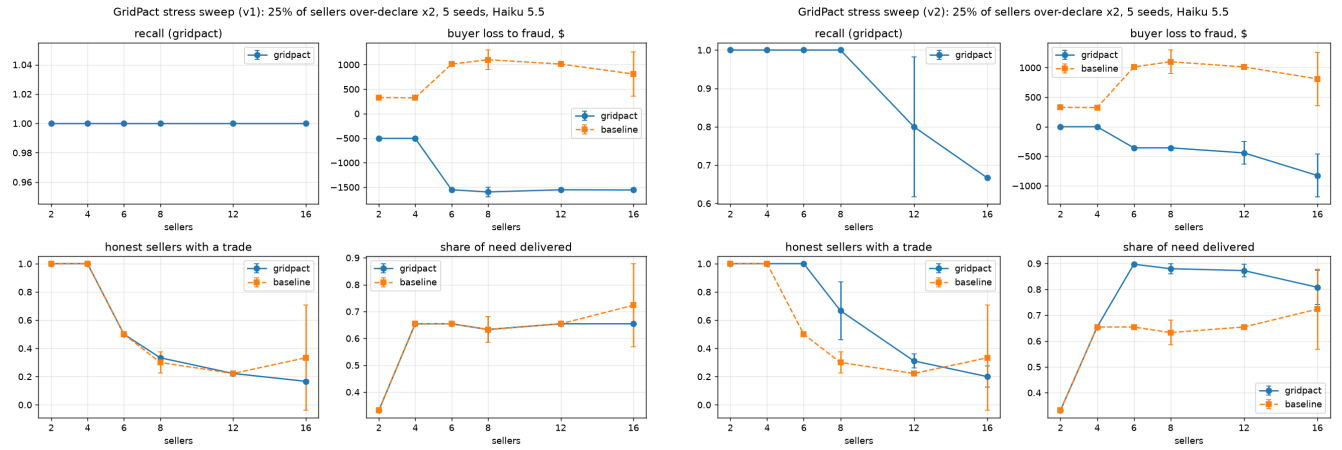
What the hard cases showed. (a) The 2 false challenges were a rounding trap: the ledger prints 6837.4 kW, the auditor's brief says 6837, so it read a 0.4 kW gap as a lie and lost its $50 bond. (b) The 1.08x liar is under the 1.10x tolerance; the auditor challenged it in 5/5 runs and lost the bond each time (brief says "exceeds nameplate", server rule is 1.1x), though settlement still charged the shortfall. (c) Told it had a side deal with a liar, the auditor challenged it anyway (5/5). (d) The lazy (2-turn) auditor still filed both challenges. (e) **stake_only has the same fraud loss as gridpact**: for one day's money the deterministic stake does the work and the LLM auditor is redundant; the auditor adds the reputation cut, the cap reset and, from v2, freed volume.

5. The limit, the one change, and the retest

Limit (stress sweep, tuning cases). Recall stayed 1.0 at every n, and no agent hit its turn cap. The system fails somewhere else: from n = 6 sellers the declared capacity exceeds the buyer's need, so the buyer fills it *on paper* from the nearest four sellers (two inflated) and never contacts the rest. Real delivery was **65% of need at n = 6, 8, 12 and 16 in both conditions**, and the share of honest sellers that trade fell from 1.0 to 0.22. GridPact refunds the buyer but the energy outcome equals the baseline.

The change (v2). A valid challenge now trims the liar's unsettled commitments to its real cap, releases the stake on the cut part, and gives the buyer one more round to re-source the released volume from sellers it has not contracted. Expected effects were written down before the retest (PREREGISTRATION.md, "Phase 3 decision"). Only gridpact was rerun, same scenarios, repetitions, model and limits. **Tuning cases** are where the problem was found; **held-out cases** are the fresh check.

set	metric (gridpact)	v1	v2
tuning stress, n=6/8/12/16	share of need delivered	0.65 / 0.63 / 0.65 / 0.65	0.90 / 0.88 / 0.87 / 0.81
tuning stress, n=6 and n=8	honest sellers trading	0.50, 0.33	1.00, 0.67
tuning stress, n=12/16	recall	1.00 / 1.00	**0.80 / 0.67** (new blind spot)
tuning, 9 scenarios	fraud loss / liar gain, $/run	-380 / -462	-57 / -139
held-out (fresh), 6 scenarios	honest sellers trading, pooled	0.833	1.000
held-out double_sell_h	share of need; honest seller trading	0.87; idle in 5/5 runs	1.00; trades in 5/5
held-out, other 5	share of need delivered	unchanged	unchanged (no spare sellers, as predicted)
all rerun scenarios	false challenges; cost per run	2/160; $0.0056	0/160; $0.0069 (+15-45% from the extra round)



Stress sweep, v1 (left) and v2 (right): benchmark/results/stress_v1.png, stress_v2.png.

Costs of the change, found only by running it. At n = 12 and 16 the re-sourced volume sometimes lands on a second liar after the audit, so that liar is never challenged (recall 0.80 and 0.67); the stake still charges the shortfall, so the buyer is not out of pocket, but the challenge layer misses it. A second audit would fix it; that is a second change and was not made. Because trimming removes the shortfall, liars pay less in penalties: lying still lost money in every run, but the gap shrank from about -$600 to -$100 per run. Delivery still falls short of 100% at n >= 6 because several honest sellers hold under 3 h of energy and run dry late in the window (simulator truth, not a lie).

6. Independent replication (fresh runs, same code and settings)

To check that nothing depends on one lucky batch, the whole benchmark was rerun from scratch with the live model: v1 from a checkout of tag **v1** (190 runs, folder `v1r`) and v2 from the v2 code (85 runs, `v2r`), about $1.2 in total, zero failed runs. Re-scoring the original logs reproduced every committed table and plot exactly (the analysis is deterministic given the logs).

quantity (gridpact)	original	replication
v1 checks P1-P6	all pass	all pass
P1 liars caught	50/50	50/50
P2 false challenges	2/160	0/160 (95% upper bound 2.3%): the false-challenge rate is real but occasional
P3 / P4 mean per run	-$538 / -$664	-$545 / -$673
P6 cost ratio	1.37x	1.40x
v2 checks; false challenges; fraud loss	pass; 0/160; -$63	pass; 0/160; -$63
v2 recall at n=12 / 16	0.80 / 0.67	0.87 / 0.73 (blind spot reproduced)
v1 share of need delivered, n=6/8/12/16	0.65 / 0.63 / 0.65 / 0.65	0.65 / 0.63 / **0.69 / 0.88**
v2 share of need delivered, n=6/8/12/16	0.90 / 0.88 / 0.87 / 0.81	0.90 / 0.90 / 1.05 / 0.84
held-out honest trading, v1 -> v2	0.833 -> 1.000	0.833 -> 1.000

The money results and the v2 effect reproduce; the 65% plateau reproduced at n = 6 and 8 but **not fully at n = 12 and 16** (the buyer sometimes reached more sellers), so the size of the v1 stress limit at large n is noisy. Values above 1.0 mean the buyer over-contracted after re-sourcing; we did not investigate that.

7. What the evidence supports, and what remains unproven

Supported (original and replication): in this setting every catchable liar was caught before settlement (100/100 across both batches), honest sellers were rarely challenged, the buyer lost nothing to fraud, lying lost money, and trust settlement paid liars in every run where the lie bit. The money outcomes are deterministic once trades exist, so the uncertainty sits in the LLM layers (P1, P2, which trades form). The v2 change improved delivery and honest trading on both tuning cases and the fresh held-out case, at the price of weaker deterrence and a new blind spot at large n.

Unproven or limited: (1) Only inflated declared capacity was tested, not lies in metered delivery or telemetry; delivery is bounded by the registry nameplate. (2) The auditor is handed the nameplates, so it compares two numbers rather than gathering evidence; a fixed rule would catch the same liars, so this does *not* show LLM auditing is needed. (3) One model (Haiku 5.5, default temperature), one day, mostly one buyer; five repetitions per case give wide uncertainty, and the 50/50 recall means "at least about 93% with 95% confidence", not 100%. Runs within a scenario are not independent samples of scenarios. (4) The held-out set shares the simulator and harness with tuning, so it checks the change, not generality. Only one held-out case (double_sell_h) had room to improve, so the held-out evidence for v2 is thin. (5) The v1 plateau at n = 12 and 16 did not fully replicate. (6) The auditor's brief and the server's 1.1x rule disagree; fixing that was not done, to keep v1 and v2 as run.

8. Evidence and how to repeat

Execution evidence (every tool call and result, ground truth, all database tables, cost, tokens, time per run): original v1 **https://github.com/nh13120/gridpact/tree/v1/benchmark/runs/v1**; original v2 **https://github.com/nh13120/gridpact/tree/v2/benchmark/runs/v2**; replication v1r **https://github.com/nh13120/gridpact/tree/main/benchmark/runs/v1r**, v2r **https://github.com/nh13120/gridpact/tree/main/benchmark/runs/v2r**. Tables: `benchmark/results/` (summary_v1, stress_v1, summary_v2, stress_v2, compare_v1_v2, summary_v1r, summary_v2r). Spec, pre-set checks and dated amendments: `benchmark/PREREGISTRATION.md`. Narrative: `benchmark/README.md`.

To repeat: follow CLAUDE.md (Python 3.13 venv, `pip install -r requirements.txt`, `.env` with an Anthropic API key, the three ENGIE parquet files in `data/`, which are sponsor data and not in the repo), then `python -m benchmark.run --condition both --scenario all --seeds 5 --parallel 4 --version v1` from tag v1, and `python -m benchmark.run --condition gridpact --version v2 --seeds 5 --scenario liar2x_of_2,liar2x_of_4,subtle115_of_4,subtle108_of_4,double_sell,dropout_of_4,lazy_auditor,stress,heldout` from tag v2; then `python -m benchmark.score / analyze / stress / compare` (full commands: PREREGISTRATION.md section 10). A run costs about half a cent and 25-40 s; the full benchmark about $1.5 against a preregistered $15 cap. Without the data, every reported number can still be regenerated from the committed logs with the score/analyze/stress/compare commands. No API keys or ENGIE time series are in the logs.